

Are Headtracking Measures an Indicator of Depressive Symptoms?

BRSM PROJECT
POSTER PRESENTATION
SP 2026

INTRODUCTION AND DATASET

This study tests whether head-tracking in **360 degree VR can indicate depressive symptoms**. Because users must move their head to explore VR scenes, movement speed may reflect psychomotor slowing linked to depression. We examined whether depression predicts head movement after accounting for anxiety and subjective experience.

Dataset : N=40 students (8 depressed, 32 non-depressed), each viewed 5 videos (200 observations).
Primary outcome: mean head-rotation speed; secondary measures: PHQ-9, GAD-7, STAI-T, PANAS pre/post and video valence/arousal/presence ratings.

ORIGINAL STUDY FINDINGS (REPLICATION TARGET)

Srivastava et al. (2025) work found a strong, anxiety-adjusted association between higher depression severity and reduced VR head movement, motivating this replication.

RESEARCH QUESTIONS (RQ1-RQ4)

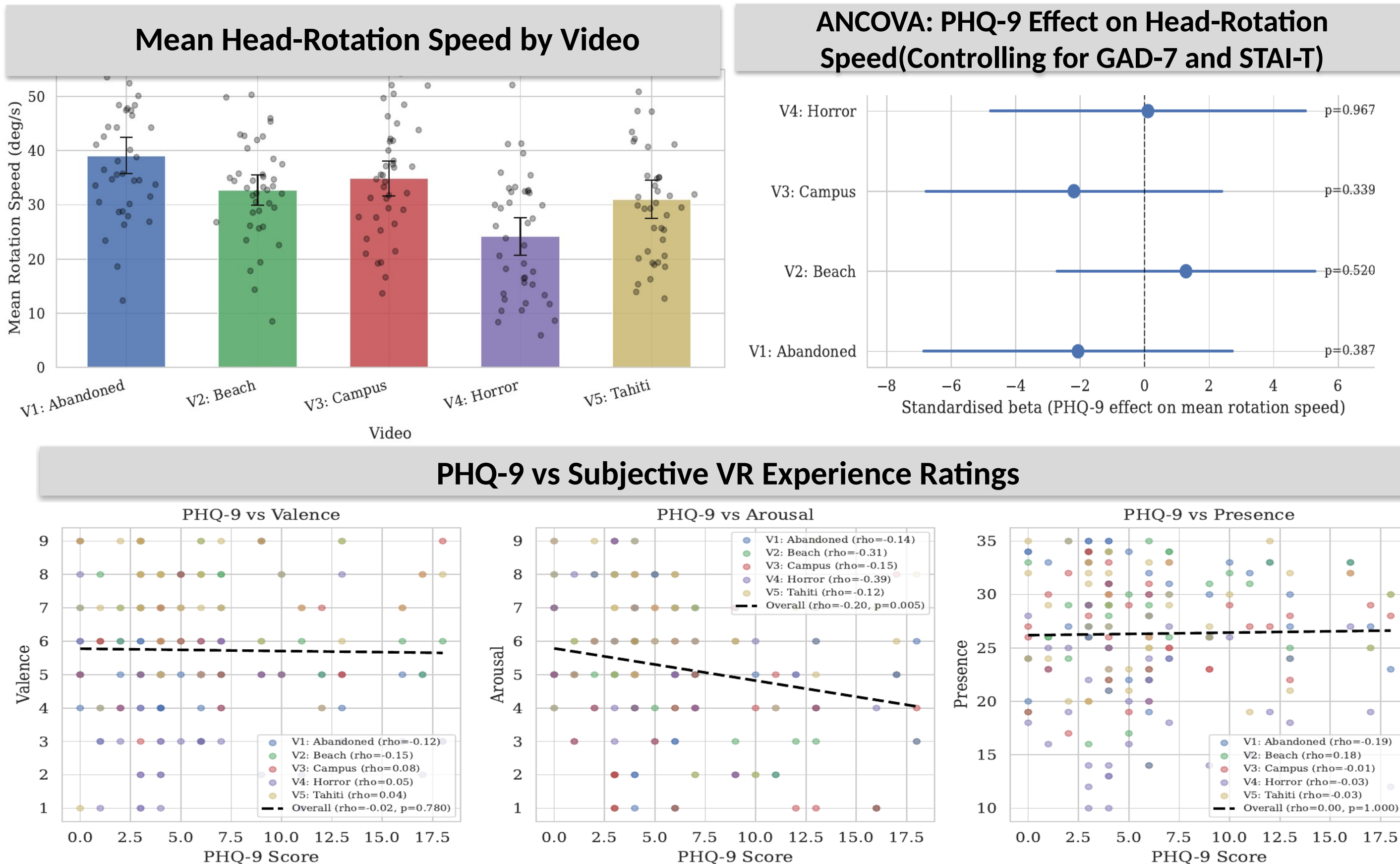
- RQ1:** Does head-rotation speed differ significantly across the five 360 degree videos, and which pairs of videos drive that effect?
- RQ2:** Does PHQ-9 score (continuous) predict mean head-rotation speed, both before and after controlling for anxiety (GAD-7, STAI-T)?
- RQ3:** Does depression influence subjective VR experience (valence, arousal, presence) and PANAS affect change?
- RQ4:** Does VR presence moderate the relationship between depression and head-movement?

METHODOLOGY

- RM-ANOVA** + Mauchly sphericity across 5 videos; Holm-corrected post-hoc paired t-tests.
- Kruskal-Wallis** + 10 pairwise Mann-Whitney U (Bonferroni-corrected) non-parametric video effect.
- Friedman's ANOVA** rank-based within-persons; Kendall's W effect size.
- Simple linear reg.** per video: PHQ-9 (continuous) -> speed; Holm + BH-FDR.
- ANCOVA:** Speed : PHQ-9(z) + GAD-7(z) + STAI-T(z). Replicates Srivastava et al.
- LMM:** Full repeated-measures; random participant intercept. ICC.
- MCC:** 30 tests (5 videos × 6 measures); Bonferroni / Holm / BH-FDR.
- Power analysis:** Post-hoc power at $\eta^2 = 0.295$ (d=1.29); MDE at 80%
- Subjective VR:** LMMs for valence, arousal, presence; Wilcoxon PANAS; Mann-Whitney.
- Moderation:** Speed ~ PHQ(z) + Presence(z) + PHQ(z)×Presence(z).

RESULTS

Analysis	Key Result	Statistic	Interpretation
Video effect on head movement	Significant difference across 5 videos	F(4,156)= 34.58, p< 0.001, eta squared g= 0.181	Video content strongly influences head-rotation speed
Post-hoc comparisons	Horror video (V4) lowest movement vs others	Holm-corrected p< 0.001 for V4 contrasts	V4 drove most between-video differences
PHQ-9 + anxiety covariates (ANCOVA)	No PHQ-9 effect after controlling GAD-7/STAI-T	All p> 0.29	Null effect persists after covariate control
Mixed model	PHQ-9 non-significant; strong person-level variance	beta= -0.55, p= 0.710; ICC= 0.694	Individual baseline movement differences dominate
Subjective arousal	Higher PHQ-9 linked to lower arousal ratings	beta= -0.44, p= 0.012	Depression related to emotional blunting
PANAS positive affect	Depressed group showed post-VR decline	Wilcoxon p= 0.008, d= -1.54	Significant reduction in positive affect in depressed group
Presence moderation	No PHQ-9 x Presence interaction	All interaction p> 0.27	Presence did not moderate PHQ-9-head speed relation



CONCLUSION

In this study, 360° VR head-tracking did not provide a robust behavioral marker of depressive symptoms. Head-rotation speed was driven primarily by video content and stable individual movement style, not by PHQ-9 severity, even after controlling for anxiety. However, depression-related effects were evident in emotional outcomes: higher PHQ-9 scores were associated with lower perceived arousal and participants in the depressed group showed a significant reduction in positive affect after VR exposure.

LIMITATIONS

- Small and imbalanced groups: only 8 people were in the depressed group versus 32 non-depressed.
- Predominantly subclinical sample: most participants were below clinical PHQ-9 cutoff.
- Large individual differences in head movement dominated the data, potentially masking depression-related effects.

CONTRIBUTION

Parv Shah: Literature review, research framing, analysis coding, visuals, interpretation, report writing.
Gaurav Patel: Literature review, coding, cross-checking, interpretation, editing report.
Hardik Kothari: Background review, methodology, statistical analysis, validation, interpretation,

REFERENCES

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